

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 14: Equation of a Line

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Forms of the equation

1. $ABCD$ is a parallelogram with vertices $A(3, 3)$, $B(0, -2)$, $C(-4, -2)$ and $D(-1, 3)$, read from a graph. (a) Find the point P where the diagonals meet. (b) Find the slopes of CB and DA and check that these sides are parallel. (c) Find the equation of the diagonal AC . **ICSE 2025, 5 marks**

SOLUTION

(a) The diagonals of a parallelogram bisect each other, so P is the mid-point of AC : $P = \left(\frac{3 + (-4)}{2}, \frac{3 + (-2)}{2} \right) = \left(-\frac{1}{2}, \frac{1}{2} \right)$.

(b) Slope of $CB = \frac{-2 - (-2)}{0 - (-4)} = 0$ and slope of $DA = \frac{3 - 3}{-1 - (-4)} = 0$. The slopes are equal, so $CB \parallel DA$.

(c) Slope of $AC = \frac{3 - (-2)}{3 - (-4)} = \frac{5}{7}$. Using $A(3, 3)$: $y - 3 = \frac{5}{7}(x - 3)$, so $7y - 21 = 5x - 15$.

Answer: (a) $P \left(-\frac{1}{2}, \frac{1}{2} \right)$ **(b) both slopes are 0** **(c)** $5x - 7y + 6 = 0$

2 Lines parallel or perpendicular to a given line

2. Which line passes through the origin and is parallel to $3x + 4y + 7 = 0$? (a) $3x + 4y + 5 = 0$ (b) $4x - 3y - 5 = 0$ (c) $4x - 3y = 0$ (d) $3x + 4y = 0$ **ICSE 2025, 1 mark**

SOLUTION

A parallel line has the same x and y coefficients: $3x + 4y + k = 0$.

It passes through $(0, 0)$, so $k = 0$.

Answer: (d)

3. The line PQ meets the positive x -axis at P , 5 units from the origin, and the negative y -axis at Q , 3 units from the origin. The line RS passes through the origin and is perpendicular to PQ . Find (a) the coordinates of P and Q (b) the equation of RS . **ICSE 2025, 3 marks**

SOLUTION

(a) $P(5, 0)$ and $Q(0, -3)$.

(b) Slope of $PQ = \frac{0 - (-3)}{5 - 0} = \frac{3}{5}$, so the slope of RS is $-\frac{5}{3}$.

RS passes through the origin: $y = -\frac{5}{3}x$.

Answer: (a) $P(5, 0), Q(0, -3)$ **(b)** $5x + 3y = 0$

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